MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
Bachelor of Technology
Academic year 2025-26

SensePath

An Intelligent Assistive Navigation System for the Visually Impaired

Submitted By:

Raja	Mechanical Engineering/1st Year
Devansh	Mechanical Engineering/1st Year
Mayank	Metallurgical Engineering/1st Year
Kamlesh	Electrical Engineering/1st Year

PROJECT GUIDE:

Dr Suresh Kumar Gawre (Electrical department)

SensePath

Real-Time Navigation Assistance for the Visually Impaired

Aim

Our aim is to design and develop SensePath which is an intelligent assistive navigation system that give real time audio signal and guidance to visually impaired people which enabling safe mobility, and also improve independence and enhanced awareness using hardware and Intelligence of inbuilt software..

Introduction

SensePath is an intelligent device that have a navigation system designed to give real time guidance with great accuracy via audio feedback.
This device use real location of the obstacles and detect them
by object recognition using sensor and the camera module also there going to be a microcontroller to Scan the surroundings environment and interpret it. The relevant information is then given to the user by the medium of voice and the Relevant information is and then that hep to enable communicated to the safe and more confident navigation. The design of the product is focused on the
Simplicity, affordability and modularity which is essential for allowing to operate both indoor and outdoor settings while
supporting future technology like GPS integration.

Problem Statement

We can see in our surrounding that the Visually impaired individuals face challenges in doing everyday tasks that people like us can do easily. Some of them are like navigating unfamiliar things then identifying objects or obstacles also reading printed text and recognizing faces these kind of struggles that affect their independence, safety, and confidence. Even though tools such as white canes, Braille systems, and voice based smartphone apps exist, but the truth is most are either limited in functionality, costly, or require human help. Many of these solutions also fail to adapt real conditions like crowded areas, moving vehicles, and also changing light path. There is a strong need for an affordable and intelligent solution that can help visually impaired people in real time enhance their movement and reduce their dependence on others for daily things.

Objective

The main objective of this project is to **develop an affordable and intelligent system** that give power and independence to visually impaired people.
This solution aims to:

- Provide real time guidance for navigation and obstacle detection.
- Enable audio based interaction for recognizing objects, surroundings, and text.
- Use accessible technology that remains cost effective for greater use.
- Reduce dependency on others by integrating smart, voice based features that simplify daily tasks.

The ultimate goal is to create a system that combines **technology, empathy, and practicality** to improve the quality of life for the visually impaired community.

Proposed Solution

The proposed solution is a **smart assistance system** designed particularly for visually impaired people to help them navigate safely and carry out daily tasks freely.

The system will combine **voice assistance, obstacle detection, and object recognition** into one compact, easy to use setup. It will use **ultrasonic sensors** to detect nearby obstacles, a **camera module** that can identifying common objects, and a **microcontroller** to process all the data and trigger real time audio feedback directly to the visually impaired person.

Users will receive **voice alerts or vibrations** about obstacles, directions, or objects in front of them. The device may also connect with a **mobile app** for advanced features such as GPS based location guidance or emergency assistance.

By integrating affordable hardware and AI based software, the proposed system will be **cost effective, user friendly, and adaptable** to both indoor and outdoor environments.

Hypothesis

The solution is expected to work because it combines the modern technologies such as ultrasonic distance sensing, computer vision, and text to voice systems into a great and affordable device. These technologies have individually demonstrated reliability in real world applications. By integrating them into a single system and providing immediate audio feedback, SensePath can significantly improve situational awareness and navigation safety for visually impaired person .

Commercialization Potential and Business Plan

SensePath has strong commercialization potential due to its low cost hardware design and growing demand for improved technologies. The system can be manufactured as a wearable and also can be manufactured portable device and distributed through educational institutions, NGOs, healthcare providers, and government programs. Future revenue models may include device sales, customized versions for institutions, and mobile application based features. The focus remains on affordability and scalability for greater and wide adoption.

Tangible Quarterly Milestones

- **Quarter 1:** System design finalization and procurement of hardware components
- **Quarter 2:** Hardware integration and basic software implementation
- **Quarter 3:** Computer vision, audio feedback, and sensor optimization
- **Quarter 4:** Testing, validation, documentation, and prototype demonstration

Final Deliverable at the End of One Year

A fully functional prototype of the SensePath assistive navigation system, accompanied by system documentation, test results, and a demonstration showcasing real time obstacle detection and audio based navigation assistance.

Components

Main Components

1. Raspberry Pi 4 Model B
   Acts as the central processing unit, handling sensor data, image processing, and decision-making while coordinating all hardware and software operations in real time.

2. Micro SD Card (16GB / 32GB)
   Used to store the operating system, application code, and data files required for system booting, processing tasks, and smooth execution of software modules.

3. Raspberry Pi Camera Module V2
   Captures real-real time visual data for object detection and recognition tasks, enabling the system to analyse surroundings using computer vision techniques.

4. Headphone / Earphone (3.5mm or USB)
   Provides audio output in the form of voice alerts and guidance, allowing users to receive real-time feedback without visual dependency.

5. Power Bank (5V, 2.5A)
   Supplies portable and stable power to the system, ensuring uninterrupted operation during outdoor use and making the setup suitable for wearable applications.

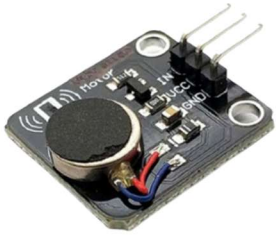
Sensors

6. Ultrasonic Sensor (HC-SR04)
Detects nearby obstacles by measuring distance using ultrasonic waves, enabling timely alerts to help users avoid collisions during navigation.

Output & Control

7. Speaker / Headphone
Delivers voice-based navigation instructions and system alerts, ensuring clear communication between the device and the user in real time.

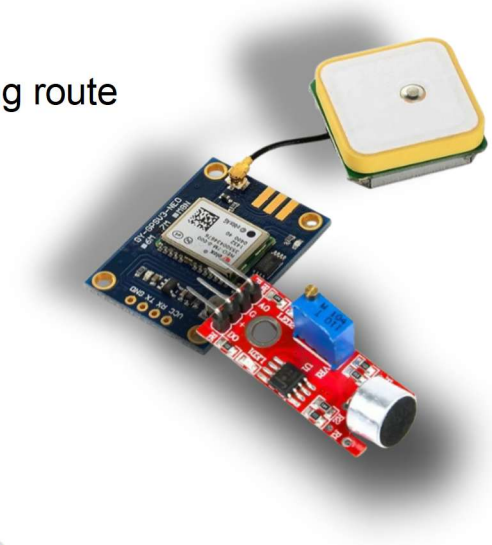
8. Push Button Switch
Allows manual control of system functions such as activation, mode switching, or emergency actions, providing simple and accessible user interaction.



9. Vibration Motor Module
Provides tactile feedback for obstacle alerts, offering an alternative output method for situations where audio feedback may not be suitable.

Additional Modules

10. GPS Module (NEO-6M)
Enables location tracking and outdoor navigation by providing real-time geographic coordinates, supporting route guidance and emergency location sharing.



11. Microphone Module (Voice Commands)
Captures voice input from the user, allowing hands-free interaction with the system through basic voice commands.

Accessories

12. Jumper Wires (Male–Female)
13. Breadboard.
14. USB Cable
15. Raspberry Pi OS
16. Wearable Case / Box

Software

17. Python 3
18. OpenCV
19. TensorFlow Lite.
20. eSpeak (Text-to-Speech).

Functionality

The proposed system will function through an integrated combination of **sensors, microcontroller, and voice feedback** mechanisms.

1. **Input Stage:**
   The system uses **ultrasonic sensors** to rapidly scan the surroundings and detect obstacles or objects in the user's path.

2. **Processing Stage:**
   The sensor data is processed by a **microcontroller or single board computer**. The system determines the distance, type, and location of detected objects.

3. **Output Stage:**
   The processed information is converted into **voice alerts or vibration signals** that guide the user safely. For instance, the system can warn: *"Obstacle ahead, two meters left."*

4. **Mobile Integration:**
   The system can be paired with a **mobile application** via Bluetooth or Wi-Fi to provide additional features like **GPS based navigation**, **location tracking**, or an **SOS button** for emergencies.

The system's overall workflow ensures **real time awareness** and **hands free operation** which is essential for allowing visually impaired people to move confidently and independently in their surroundings.

Expected Outcomes

The proposed smart assistance system is expected to deliver tangible benefits that enhance the daily live of a visually impaired person.

Key outcomes include:

- **Improved Mobility:** Users can navigate safely in both indoor and outdoor environments without constant human assistance.

- **Enhanced Independence:** The system allows visually impaired individuals to perform day-to-day activities with greater confidence.
- **Affordable Accessibility:** Designed with low-cost components to ensure affordability for all sections of society.
- **Real Time Feedback:** Immediate audio or vibration alerts increase situational awareness and reduce the risk of accidents.
- **Technological Empowerment:** Encourages inclusivity by making technology practical and accessible to people with disabilities.

This solution will ultimately **bridge the gap between disability and independence**, aligning with the broader goal of creating a more inclusive, empathetic, and technology driven society.

Future Scope

The project has strong potential for further development and real world implementation. Future enhancements can include:

- **AI and Computer Vision Integration:** Adding camera based object and face recognition using machine learning models to identify people, signals, or obstacles more accurately.
- **GPS and Cloud Connectivity:** Real time tracking, route planning, and data delivery with mobile devices for enhanced outdoor navigation.
- **Voice Command System:** Allowing users to interact with the device through natural voice commands for greater convenience.
- **Smartphone App Expansion:** Integrating accessibility features like emergency calling, location sharing, and health monitoring.
- **Miniaturization and Wearability:** Evolving the design into a compact wearable such as smart glasses, wristbands, or a clip on module.

With continuous improvement, the project can evolve into a **comprehensive assistive ecosystem** that empowers visually impaired individuals to live more independently and safely.

Additional Information

Affordability, modularity, and practical usability are highly valued in the proposed project SensePath, guaranteeing that the solution is not restricted to a lab prototype but has a definite potential for real world implementation. Because the system is built with widely accessible and reasonably priced hardware components, it can be widely adopted, particularly in settings with limited resources.

The project's modular architecture, which makes it possible to upgrade or replace specific parts like sensors, vision modules, and feedback mechanisms without completely redesigning the system, is one of its main advantages. Future improvements, such as sophisticated AI-based object recognition, voice-command interaction, GPS-enabled navigation, and miniaturization into wearable form factors like smart glasses or clip-on devices, are made possible by this flexibility.

The project also encourages interdisciplinary learning by integrating concepts from electronics, embedded systems, computer vision, and human centered design. Beyond academic value, SensePath has strong social relevance and aligns with national and global goals of inclusivity and assistive technology development. With further refinement, the system has the potential to evolve into a scalable assistive solution that benefits visually impaired individuals while contributing meaningfully to research, innovation, and societal impact.